MAT 3007 – Optimization

Introduction to Optimization

Lecture 02

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Repetition

General forms of optimization problems:

$$\text{minimize}_{x\in\mathbb{R}^n}\ f(x) \quad \text{subject to} \quad x \in \Omega \tag{1}$$

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad \forall\ i = 1, ..., m, \\ & h_j(x) = 0, \quad \forall\ j = 1, ..., p. \end{aligned}$$

Three main components in optimization problems:

- Decision.
- Objective.
- Constraints.

Vectors and Norms

Vectors and Norms - I

A point or vector $x \in \mathbb{R}^n$ is represented by a column vector

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{with components} \quad x_i \in \mathbb{R}, \quad \forall\, i.$$

The transposed $x^\top$ is the row vector $x^\top = (x_1, x_2, \cdots, x_n)$.

For $x, y \in \mathbb{R}^n$, we define the Euclidean inner product by

$$\langle x, y \rangle := \sum_{i=1}^n x_i y_i = x^\top y.$$

The inner product has the following properties:

- Positivity: $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ if and only if $x = 0$,
- Symmetry: $\langle x, y \rangle = \langle y, x \rangle$,
- Linearity: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$,

for all $\alpha, \beta \in \mathbb{R}$ and $x, y, z \in \mathbb{R}^n$.

Vectors and Norms - II

The Euclidean norm of a point $x \in \mathbb{R}^n$ is defined as:

$$\|x\|_2 = \|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^\top x} = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}.$$

The Euclidean norm satisfies the following properties for all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- Positivity: $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$.
- Homogeneity: $\|\alpha x\| = |\alpha| \|x\|$.
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$.
- Cauchy-Schwarz inequality: $|\langle x, y \rangle| \leq \|x\| \|y\|$.

$$\|x+y\|^2 = \langle x+y, x+y \rangle$$
$$= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2$$
$$= \|x\|^2 + \|y\|^2 + 2\langle x, y \rangle$$
$$\leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|$$
$$= (\|x\| + \|y\|)^2$$

Other norms:

- ℓ_1-norm: $\|x\|_1$, ℓ_p-norm: $\|x\|_p = [\sum_{i=1}^n |x_i|^p]^{1/p}$, maximum norm $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$, ...

Sets

Supremum and Infimum

The supremum of a nonempty set $\Omega \subset \mathbb{R}$ is the smallest scalar y such that

$$y \geq x \quad \text{for all} \quad x \in \Omega.$$

The infimum of a set $\Omega \subset \mathbb{R}$ is the largest scalar y such that

$$y \leq x \quad \text{for all} \quad x \in \Omega.$$

If $\sup \Omega \in \Omega$ ($\inf \Omega \in \Omega$) then $\sup \Omega = \max \Omega$ ($\inf \Omega = \min \Omega$).

Examples:

- $\sup\{1/n : n \geq 1\} = ?, \quad \inf\{1/n : n \geq 1\} = ?$ 0
- $\sup\{n : n \geq 1\} = ?, \quad \inf\{n : n \geq 1\} = ?$

why ?

① $0 \leq \frac{1}{n}, \forall n \geq 1$

② $\forall \varepsilon > 0$, we can find $n \geq 1$ s.t. $\varepsilon > \frac{1}{n}$.

Basic Terminologies and Formal Definitions

Basic Terminologies

$$\text{minimize}_{x\in\mathbb{R}^n}\ f(x) \quad \text{subject to} \quad x \in \Omega \tag{2}$$

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad \forall\ i = 1, ..., m, \\ & h_j(x) = 0, \quad \forall\ j = 1, ..., p. \end{aligned}$$

- Feasible point/Feasible solution: A decision x that satisfies all constraints.
- Domain/Feasible set/Feasible region Ω: The set of all feasible points.
- Optimal value: The infimum of f over the feasible set. We denote it as f^*. If there exists a feasible solution x^* such that $f(x^*) = f^*$, we say f^* is attainable.

An Example – Attainable and Not Attainable

Attainable: $\min_x f(x) := x^2$

s.t $x \in [0, 2]$

since $f^* = 0$ and $f(0) = 0$.

Not attainable: $\min_x f(x) = x^2$

s.t $x \in (0, 2]$

Not attainable: $\min_x f(x) = \frac{1}{x}$

s.t $x \in [1, \infty)$

Formal Definition: Local Minimizer

For $\varepsilon > 0$ and $x \in \mathbb{R}^n$ we define the neighborhood

$$B_\varepsilon(x) = \{y \in \mathbb{R}^n : \|x - y\| < \varepsilon\}.$$

The point $x^* \in \mathbb{R}^n$ is said to be a:

- Local minimizer: A feasible decision variable that achieves an objective value that is no larger than that of any other feasible point in a sufficiently small neighborhood. Mathematically, $x^* \in \Omega$ and there exists $\varepsilon > 0$ such that $f(x) \geq f(x^*)$ for all $x \in \Omega \cap B_\varepsilon(x^*)$.
- Strict local minimizer: A feasible decision variable that achieves an objective value that is strictly smaller than that of any other feasible point in a sufficiently small neighborhood. Mathematically, $x^* \in \Omega$ and there exists $\varepsilon > 0$ such that $f(x) > f(x^*)$ for all $x \in (\Omega \cap B_\varepsilon(x^*))\backslash\{x^*\}$.

Formal Definition: Global Minimizer

- Global minimizer: A feasible decision variable that achieves an objective value that is no larger than that of any other feasible point ~~in a sufficiently small neighborhood~~. Mathematically, $x^* \in \Omega$ and $f(x) \geq f(x^*)$ for all $x \in \Omega$ ~~$\cap B_\varepsilon(x^*)$~~.
- Global local minimizer: A feasible decision variable that achieves an objective value that is strictly smaller than that of any other feasible point ~~in a sufficiently small neighborhood~~. Mathematically, $x^* \in \Omega$ and $f(x) > f(x^*)$ for all $x \in (\Omega$ ~~$\cap B_\varepsilon(x^*)$~~ $)\backslash\{x^*\}$.

Question: The infimum of f is f^*. If there exists a feasible solution x^* such that $f(x^*) = f^*$, we say f^* is attainable. Is this x^* a global minimizer?

Yes. by definition of f^*.

Remarks and Comments

- global minimizer $\equiv$ global solution $\equiv$ optimal solution.
- We say the objective value is unbounded or the optimal value is $-\infty$, if for every $B \in \mathbb{R}$ there exists $x \in \Omega$ with $f(x) \leq B$.
- The definition for maximizer is identical, just change : $\geq / >$ $\to \leq / <$.

- The following optimization problems are equivalent:

$$\max_x \ f(x) \quad \text{s.t.} \quad x \in \Omega \quad \text{and} \quad \min_x -f(x) \quad \text{s.t.} \quad x \in \Omega.$$

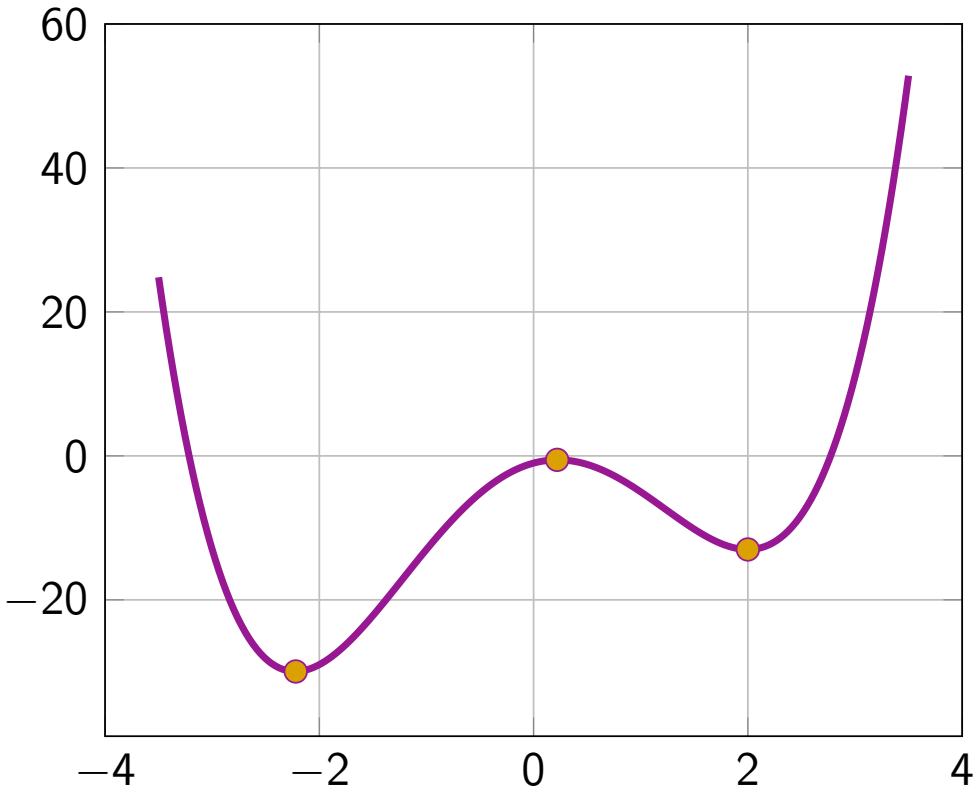
  Every loc. maximum of f on Ω is a loc. minimum of $-f$ on Ω.

$\rightsquigarrow$ We often concentrate on minimization problems.

Example: Minimizer

We consider the unconstrained problem

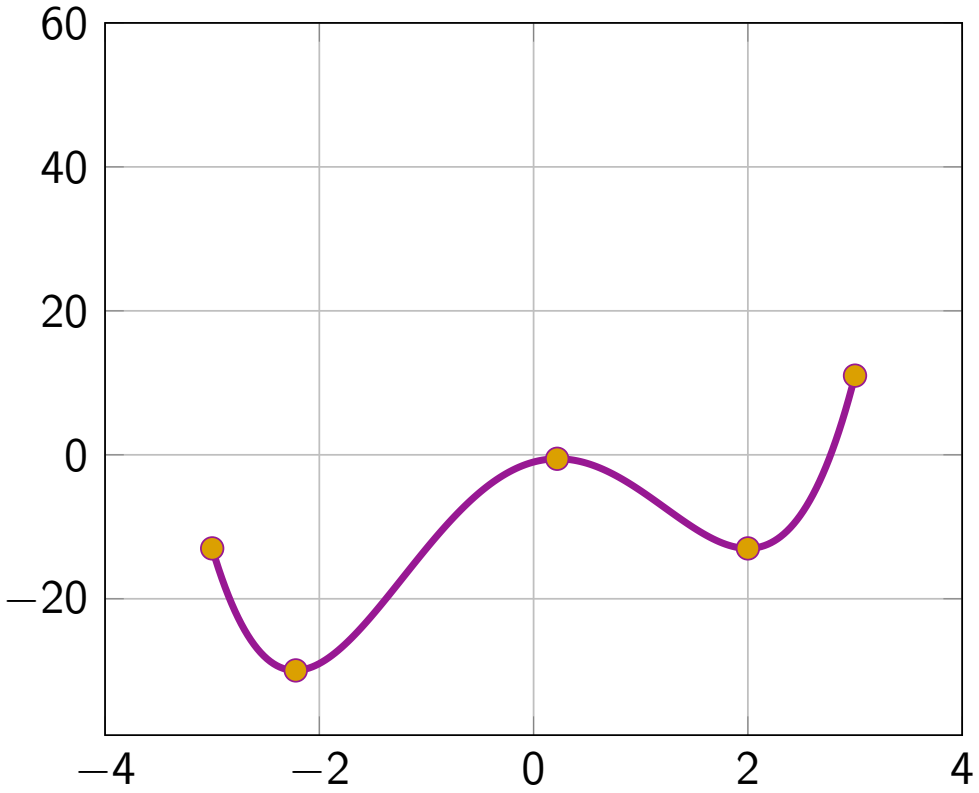
$$\text{minimize}_{x\in\mathbb{R}}\ f(x) := x^4 - 9x^2 + 4x - 1.$$



Example: Minimizer

Let us consider the constrained problem

$$\begin{aligned} \text{minimize}_{x\in\mathbb{R}} \quad & f(x) := x^4 - 9x^2 + 4x - 1. \\ \text{subject to} \quad & x \in [-3, 3]. \end{aligned}$$

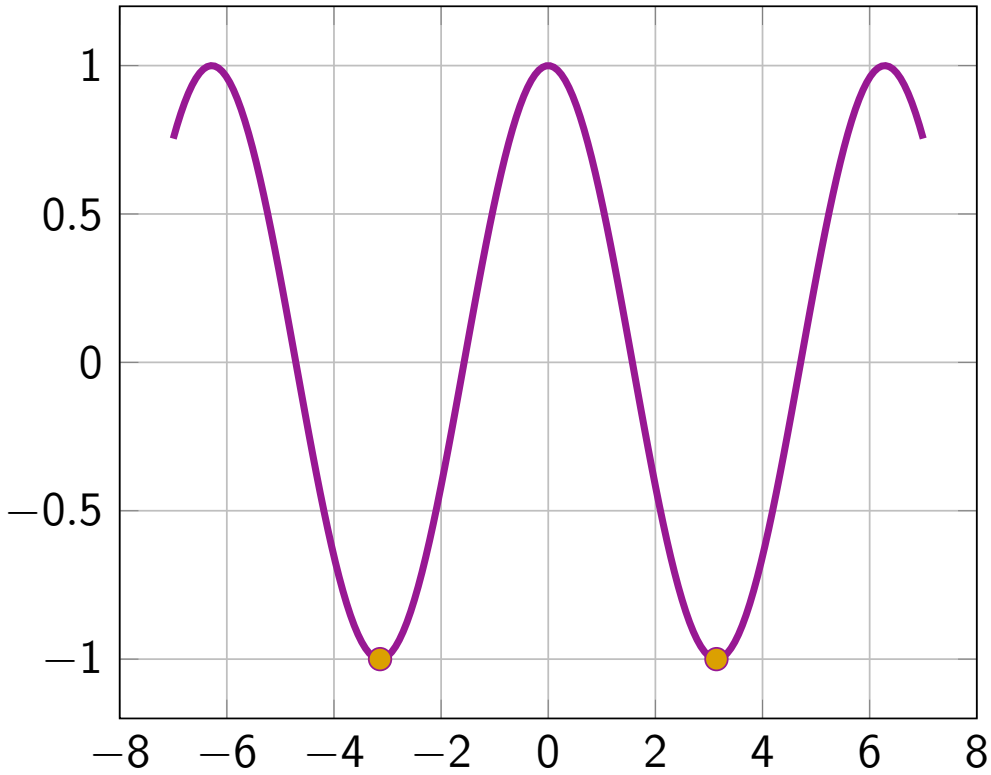


Example: Minimizer

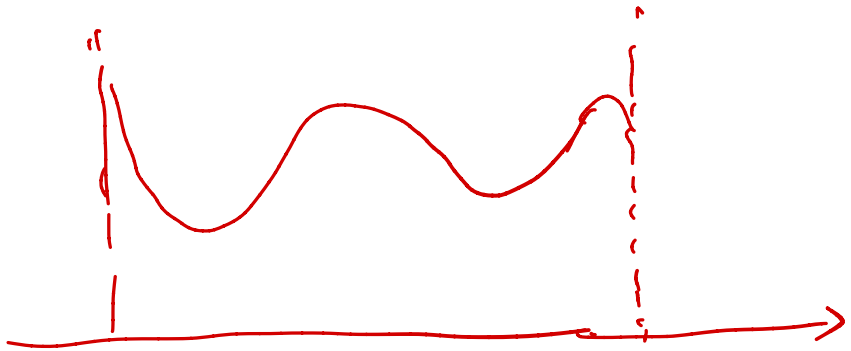
We consider the unconstrained problem

$$\text{minimize}_{x\in\mathbb{R}}\ f(x) := \cos(x).$$

- All points in $S = \{(2k+1)\pi : \pi \in \mathbb{Z}\}$ are global minima.
- None of the global minima is strict.



Example: How many local solutions are here?



How many local minimizer?

<A> 1

<B> 2

<C> 3 ✓

<D> 4

Classification of Optimization Problems

Optimization Problem:

$$\begin{aligned} \text{minimize}_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad \forall\ i = 1, ..., m, \\ & h_j(x) = 0, \quad \forall\ j = 1, ..., p. \end{aligned}$$

- Unconstrained optimization: If $\Omega = \mathbb{R}^n$ or $m = p = 0$. Otherwise: constrained optimization.
- Linear optimization (LP): Constraints and objective are linear.
- Nonlinear optimization (NLP): Either some of the constraints or the objective function is nonlinear.
- Integer/Discrete optimization (IP): Some of the decision variables have to be integers ($x \in \mathbb{Z}^n$).
- Continuous optimization: All the decision variables can take continuous values.

Classifications

- Constrained vs Unconstrained.
- Linear vs Nonlinear.
- Continuous vs Discrete.

- By default, when we talk about an optimization problem, we assume it is a continuous problem, unless we explicitly state that it is discrete.

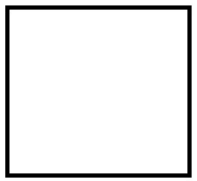
Example : Maximum Area

Example 1: Maximum Area Problem

You have 80 meters of fencing and want to enclose a rectangular shape (for example a yard) – as large as possible (in area).

- How should you do it?





Question: Constrained or unconstrianed? No Answer.

Remarks

The above classifications are based on the structure of f, g, h, and of the feasible set Ω:

- Sometimes an NLP can be equivalently transformed to an LP.
- Sometimes an IP can be equivalently transformed to a continuous optimization problem.

Linear optimization is the most well-studied and the easiest optimization problem:

- Nonlinear optimization and integer optimization could be significantly harder than LP.
- Therefore, in many cases, people strive to find LP formulations for problems.

Modeling: How to Formulate an Optimization Problem?

Modeling

Modeling is extremely important:

- Finding a good optimization model is at least half way in solving the problem.
- In practice, we are not given a mathematical formulation — typically the problem is described verbally by a specialist within some domain. It is extremely important to be able to convert the problem into a mathematical formulation.
- Modeling is not trivial: Sometimes, we not only want to find a formulation, but also want to find a good formulation.

- We now discuss some examples about modeling a problem as an optimization problem.

Formulating Optimization Problems

The golden rule: Identify and formulate the three components:

- Decision $\rightsquigarrow$ Decision variables.
- Objective $\rightsquigarrow$ Objective function.
- Constraints $\rightsquigarrow$ Constraint functions: equalities/inequalities.

Now let's make it work!

Maximum Area Problem: Revisited

You have 80 meters of fencing and want to enclose a rectangular area – as large as possible. How should you do it?

- Decision variable: the length ℓ and width w of the area.
- Objective: maximize the area: ℓw.
- Constraints: the total length of the fence: $2\ell + 2w \leq 80$.

Therefore, the optimization problem can be written as:

$$\begin{aligned} \text{maximize}_{\ell,w} \quad & \ell w \\ \text{subject to} \quad & 2\ell + 2w \leq 80 \end{aligned}$$

Maximum Area Problem: Revisited

You have 80 meters of fencing and want to enclose a rectangular area – as large as possible. How should you do it?

- Decision variable: the length ℓ and width w of the area.
- Objective: maximize the area: ℓw.
- Constraints: the total length of the fence: $2\ell + 2w \leq 80$.

Therefore, the optimization problem can be written as:

$$\begin{aligned} \text{maximize}_{\ell,w} \quad & \ell w \\ \text{subject to} \quad & 2\ell + 2w \leq 80 \\ & \ell, w \geq 0 \end{aligned}$$

Question: Constrained or unconstrained? Linear or nonlinear? Continuous or discrete?

Production Problem: Revisited

A company needs to decide the amount of each product to produce.

	Steel	Iron	Copper	Profit
Alloy 1	1	0	1	\$1
Alloy 2	0	2	1	\$2
Resources	100	200	150	

Decision variables:

- x_1: the amount of alloy 1 to produce.
- x_2: the amount of alloy 2 to produce.

Objective function:

- $f(x) = x_1 + 2x_2$

Constraints?

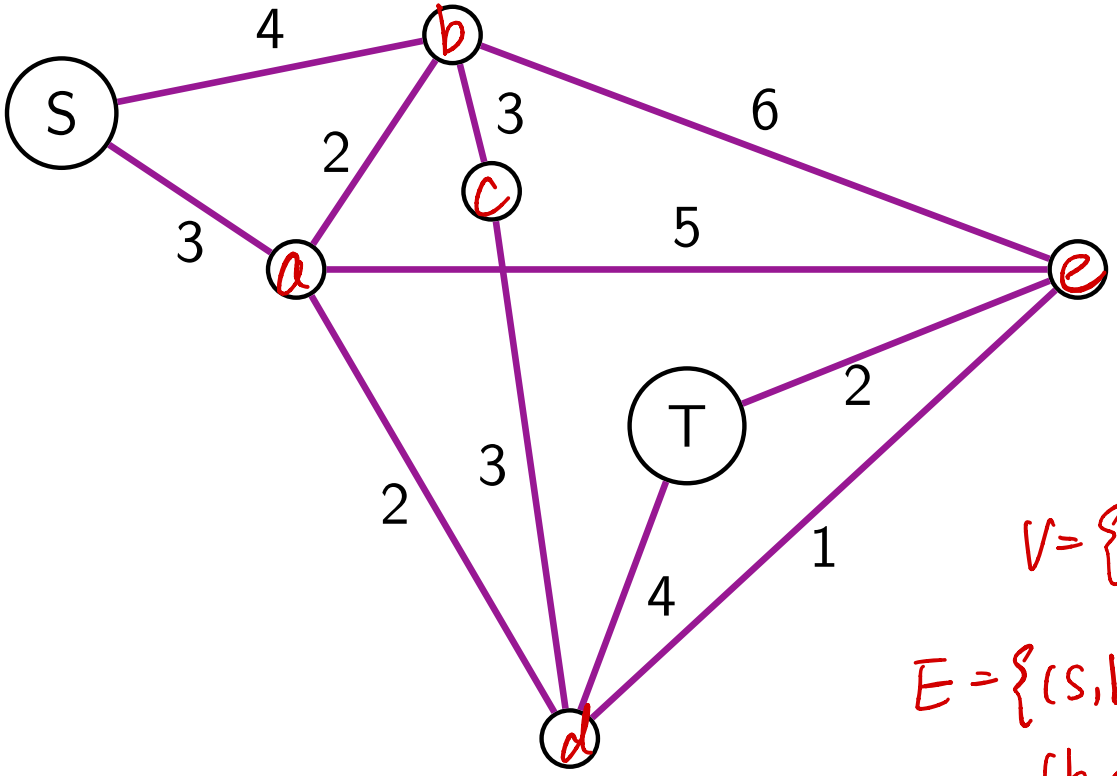
Optimization Model

$$\begin{aligned} \text{maximize} \quad & f(x) = x_1 + 2x_2 \\ \text{subject to} \quad & x_1 \leq 100 \\ & 2x_2 \leq 200 \\ & x_1 + x_2 \leq 150 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- This is a linear optimization problem.
- We call the numbers $(1, 2, 100, 150, 200,$ etc.) the coefficients of the optimization problem.

Modeling: Shortest Paths

Shortest Path Problem



$V = \{a, b, c, d, e, S, T\}$

$E = \{(S,b), (b,S), (a,b), (b,a) \cdots\}$

What is the shortest path from s to t? How to formulate it as an optimization problem?

- This is called the shortest path problem.

Some notations: Set of nodes: V. Set of edges: $E \subset V \times V$. The distance between node i and node j is w_{ij}.

Model Shortest Path Problem using Optimization

For each edge $(i,j) \in E$, define

$$x_{ij} = \begin{cases} 1 & \text{if we use edge } (i,j), \\ 0 & \text{otherwise.} \end{cases}$$

An optimization model:

$$\begin{aligned} \text{minimize} \quad & \sum_{(i,j)\in E} w_{ij} x_{ij} \\ \text{subject to} \quad & x_{ij} \in \{0,1\} \qquad \forall\, (i,j) \in E \\ & \sum_j x_{sj} = 1 \\ & \sum_j x_{jt} = 1 \\ & \sum_j x_{ij} = \sum_j x_{ji}, \quad \forall i \neq s, t. \end{aligned}$$

→ total length

→ start from S

→ end at T

→ form a path.

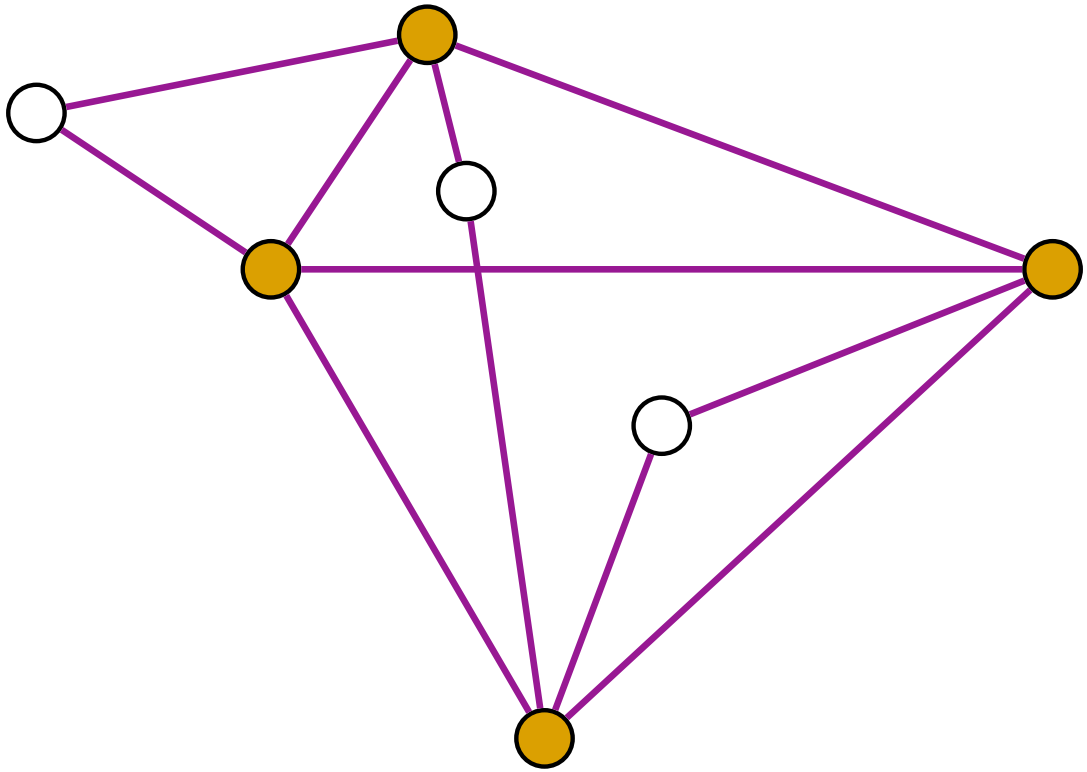
What category does this optimization problem belong to?

- Constrained, linear, integer.

Modeling: Vertex Cover Problem

Another Graph Problem: Vertex Covering

Task: Given a graph with nodes V and edges E, find the smallest set of vertices that touch every edge of the graph.



Optimization Model for Vertex Cover Problem

Define the following decision variables:

$$x_i = \begin{cases} 1 & \text{if we choose vertex } i, \\ 0 & \text{otherwise.} \end{cases}$$

An optimization model can be written as:

$$\begin{aligned} \text{minimize} \quad & \sum_i x_i \\ \text{subject to} \quad & x_i + x_j \geq 1 \quad \forall\, (i,j) \in E \\ & x_i \in \{0,1\} \quad \forall\, i \in V. \end{aligned}$$

- This is an integer (linear) optimization problem.
- Many graph/network problems can be modeled as optimization problems.

Modeling: Support Vector Machines

Support Vector Machines – Setup

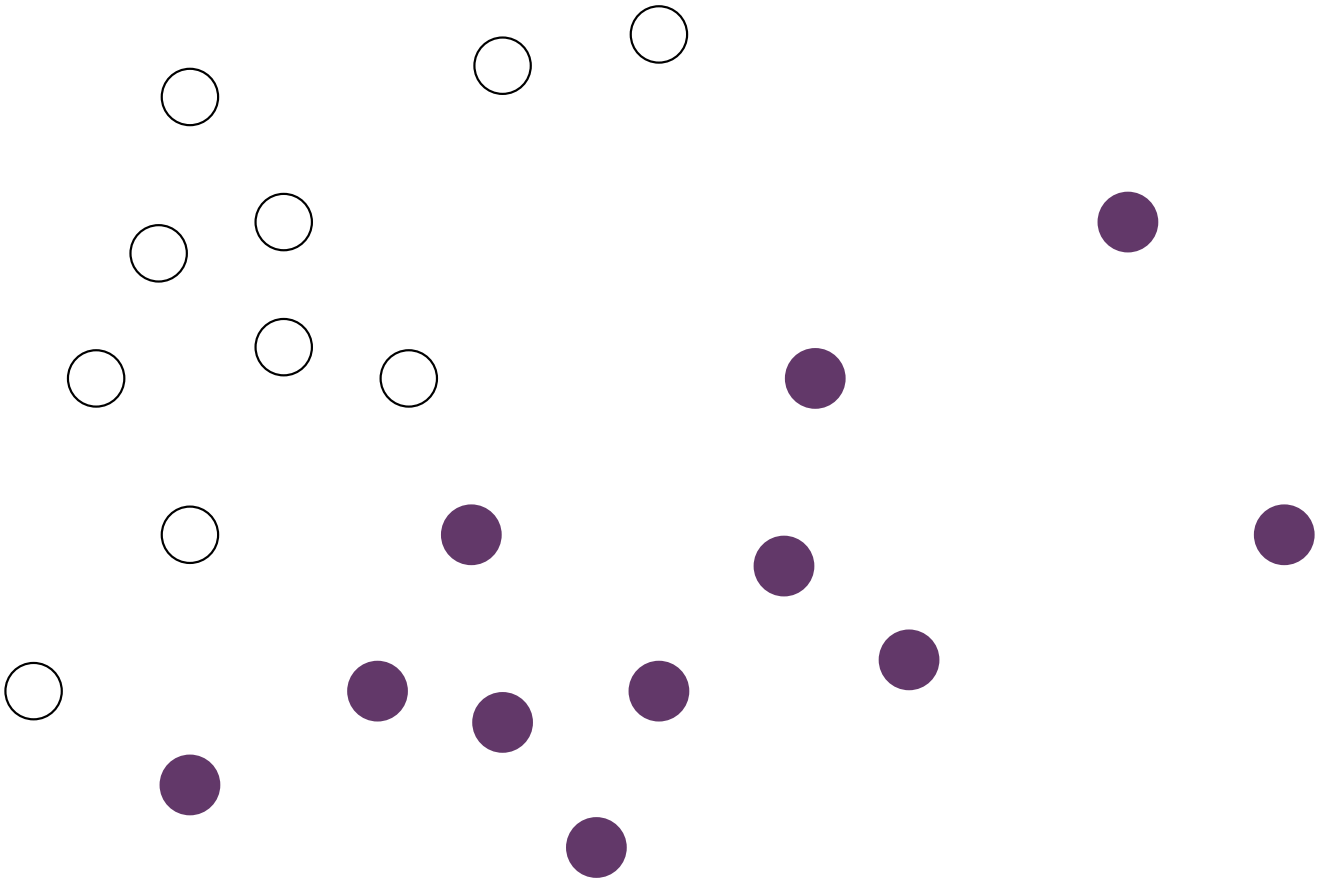
General Setup:

- Given: m objects represented by vectors $x_1, x_2, ..., x_m \in \mathbb{R}^n$ with labels $y_i \in \{-1, 1\}$.
- There are two classes, A and B.The object i belongs to class A if $y_i = 1$ and belongs to B if $y_i = -1$.
- Goal: Learn a function $\ell : \mathbb{R}^n \to \mathbb{R}^n$ based on the training sample $(x_1, y_1), ..., (x_m, y_m)$, so that when we see a new object z, the function $\ell(z)$ predicts its label/class.

- Example: Blind taste – Does the taste determine the color?

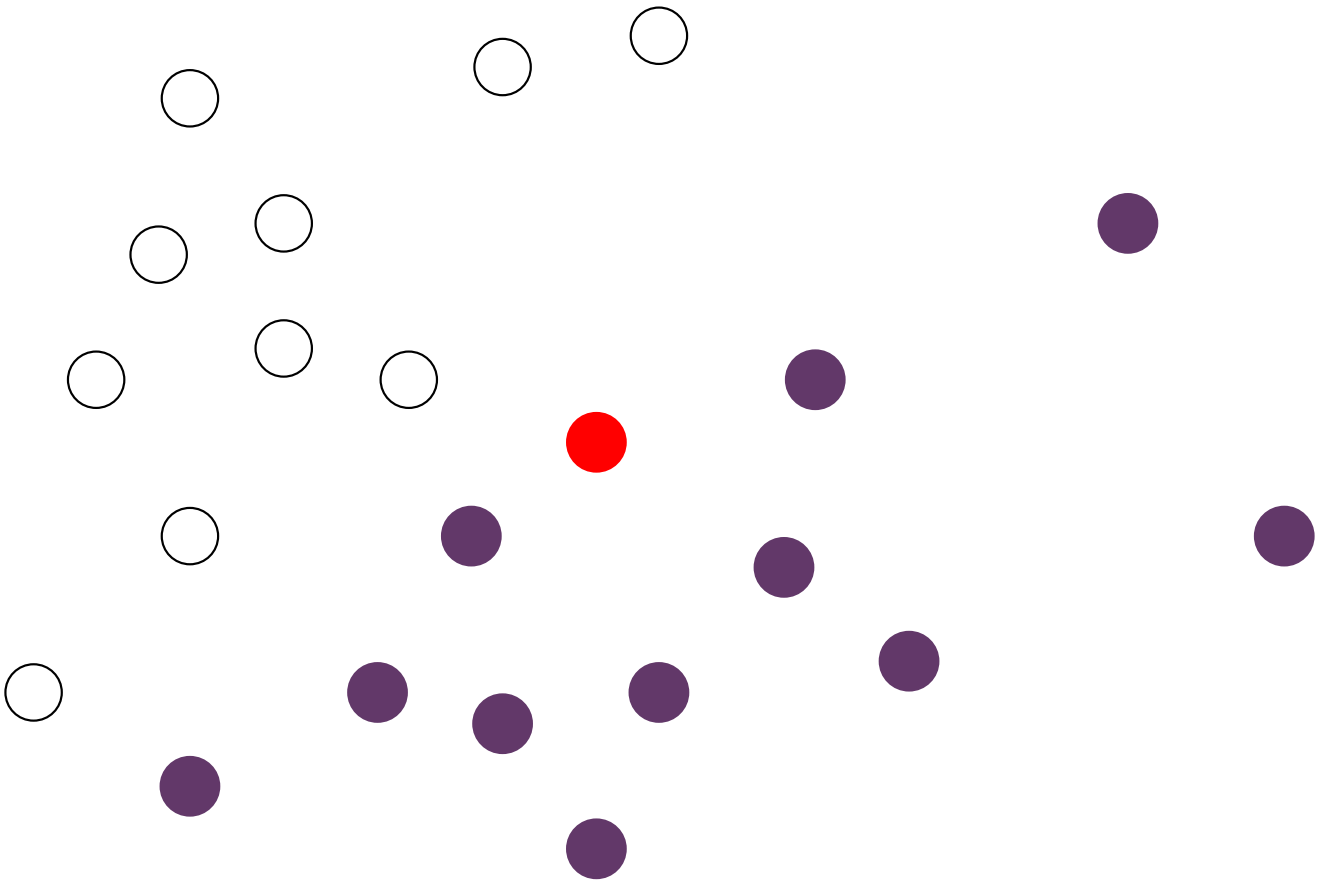
$$x_i \equiv \begin{bmatrix} \text{juicy/fresh} \\ \text{body} \\ \text{acidity} \\ \vdots \end{bmatrix}, \quad y_i = \begin{cases} +1 & \text{white wine,} \\ -1 & \text{red wine.} \end{cases}$$

Support Vector Machines: Illustration



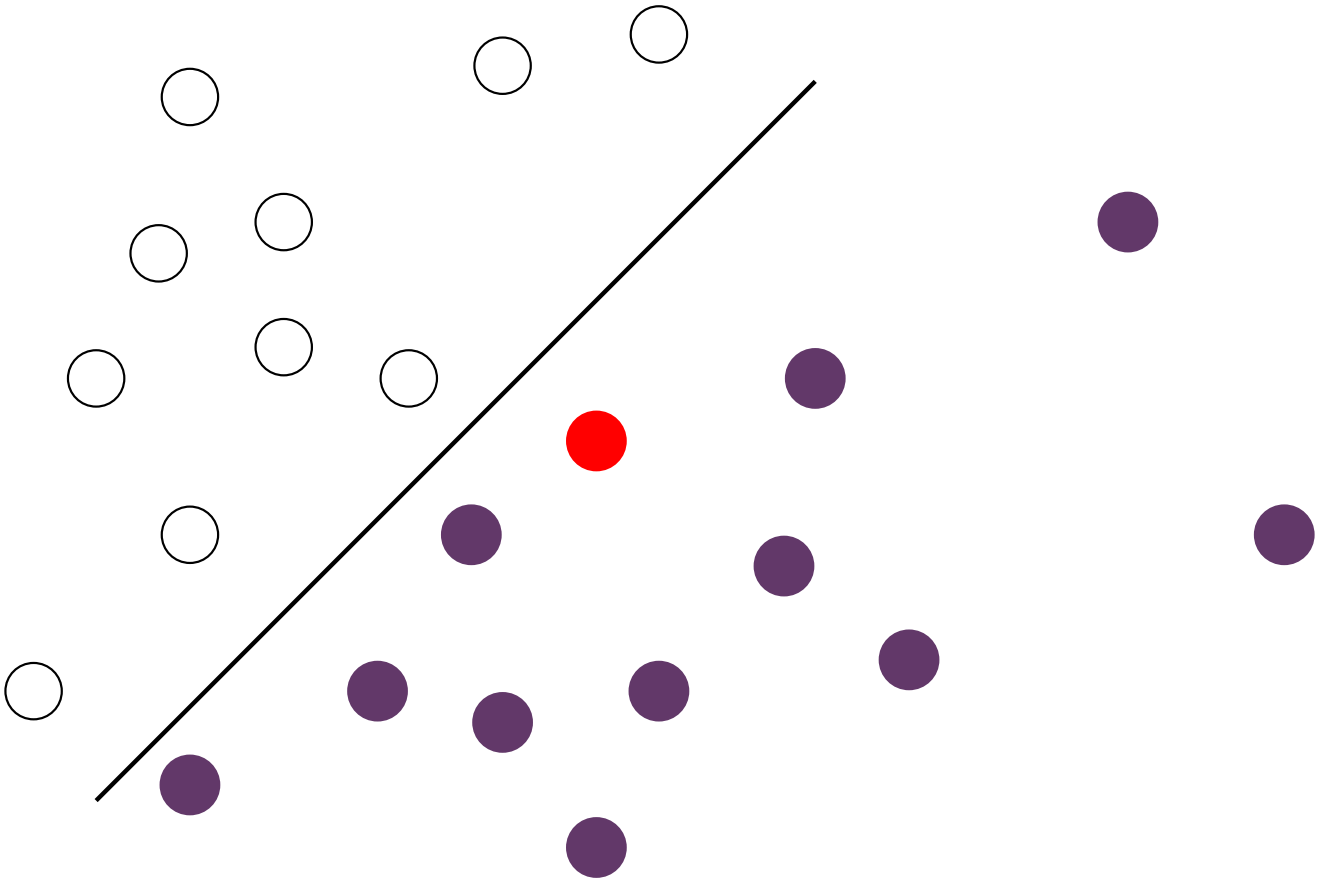
- Consider two labeled sets of points (white and purple).

Support Vector Machines: Illustration



- Consider two labeled sets of points (white and purple).
- $\rightsquigarrow$ Question: Can we predict the label of a newly added point?

Support Vector Machines: Illustration



- Separate the two data sets with a hyperplane!

Linear Classification

Assume: there exists a hyperplane $\tilde{\ell}(x) := \tilde{w}^\top x + \tilde{b}$ defined by $(\tilde{w}, \tilde{b})$ that separates the datapoints such that:

$$y_i = \begin{cases} +1 & \text{if } \ell(x_i) > 0, \\ -1 & \text{if } \ell(x_i) \leq 0. \end{cases}$$

Then there must exist a hyperplane $\ell(x) := w^\top x + b$ defined by (w, b) such that:

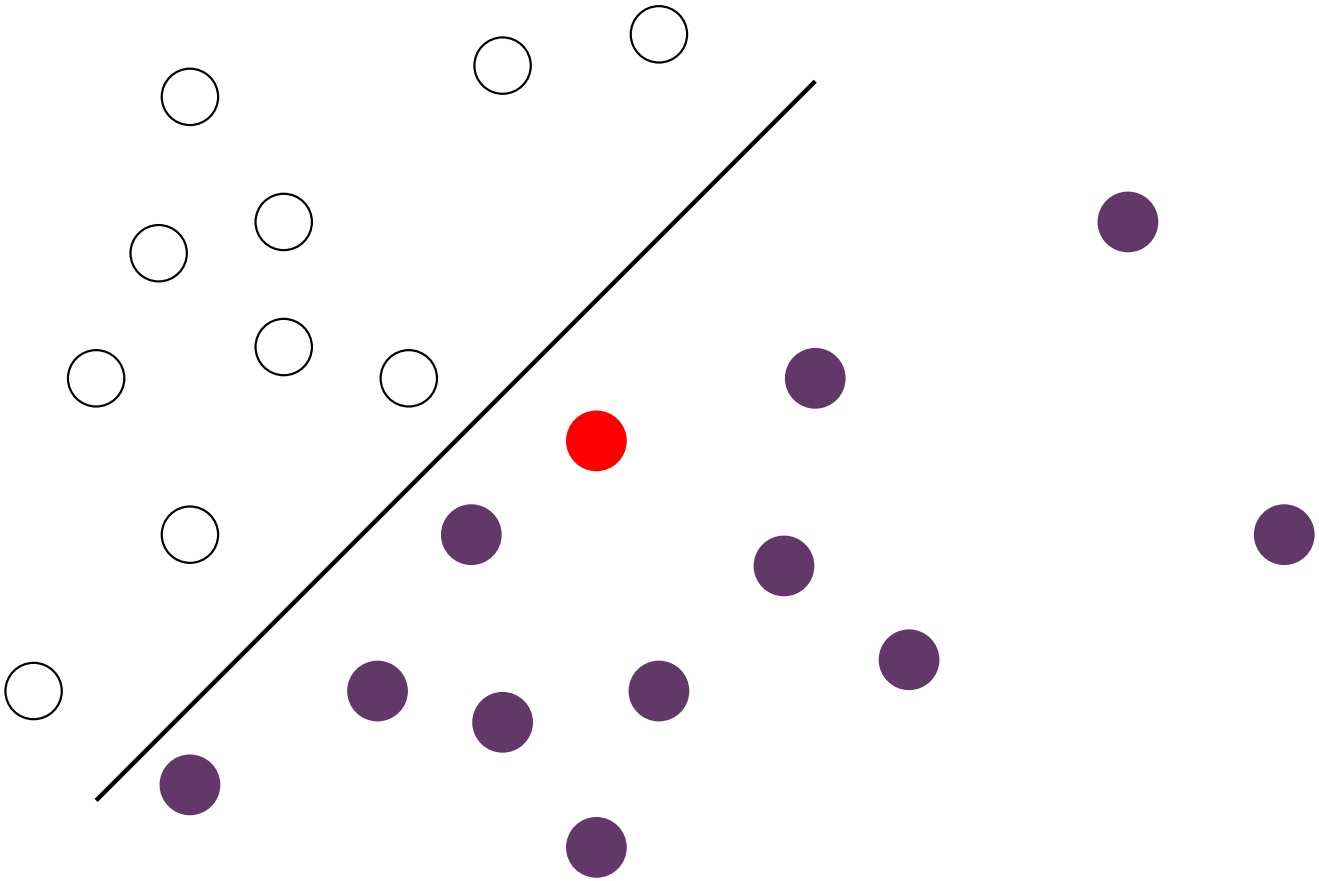
$$y_i = \begin{cases} +1 & \text{if } \ell(x_i) \geq +1, \\ -1 & \text{if } \ell(x_i) \leq -1, \end{cases} \quad \forall\ i = 1, ..., m.$$

Goal: find such (w, b). The associated optimization problem:

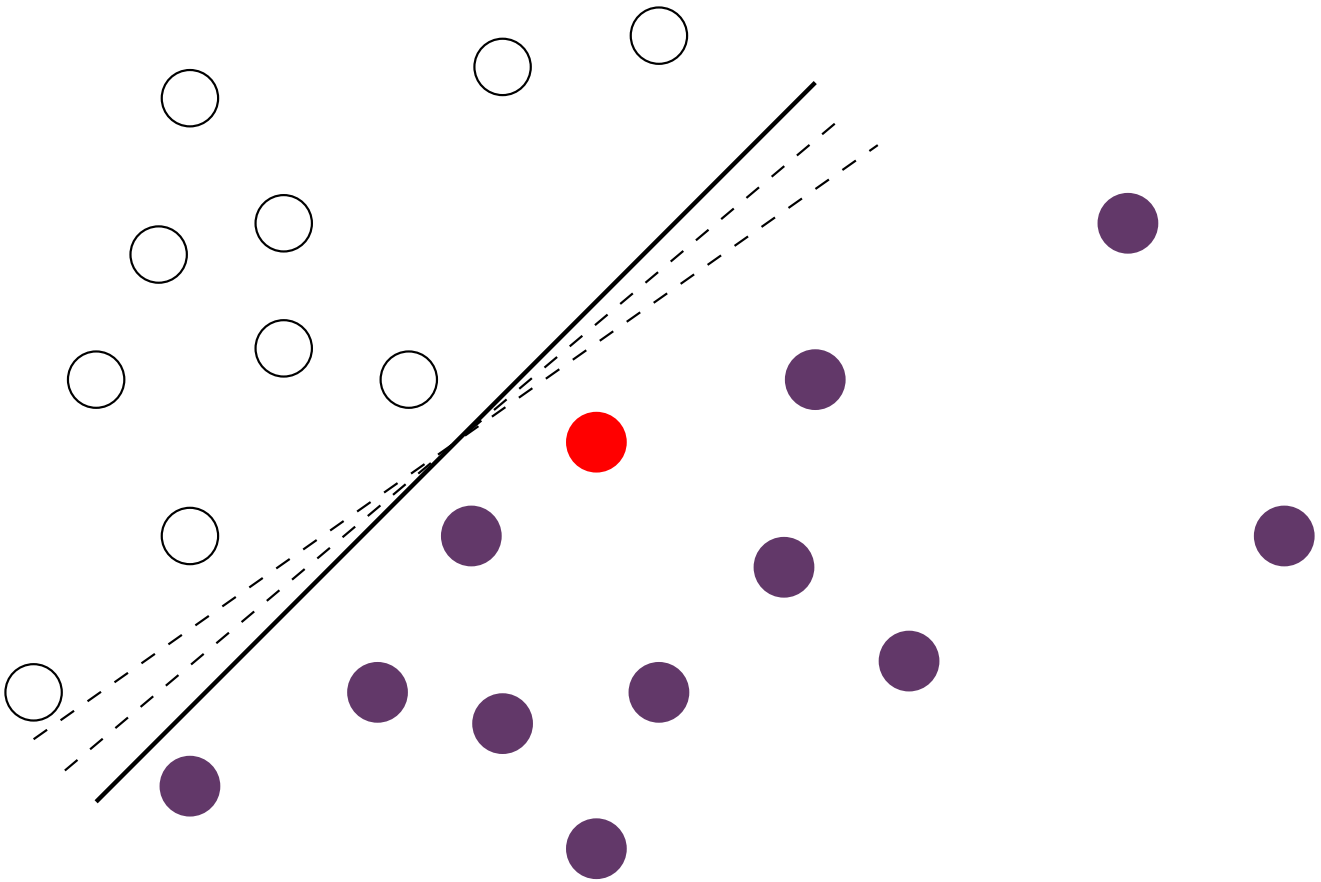
$$\text{minimize}_{w,b}\ 0 \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\ i.$$

- This problem is a feasibility problem: feasibility problems are a special kind of optimization problem.

Support Vector Machines: Illustration



Support Vector Machines: Illustration



- Problem: The separating hyperplane might not be unique!

Support Vector Machines: Model Improvement

- Select the "best" hyperplane to separate the two groups.
- Distance between the hyperplanes $\{x : w^\top x + b = 1\}$ and $\{x : w^\top x + b = -1\}$ is $2/\|w\|$ (why?).

Maximize the possible margin (distance between the datasets):

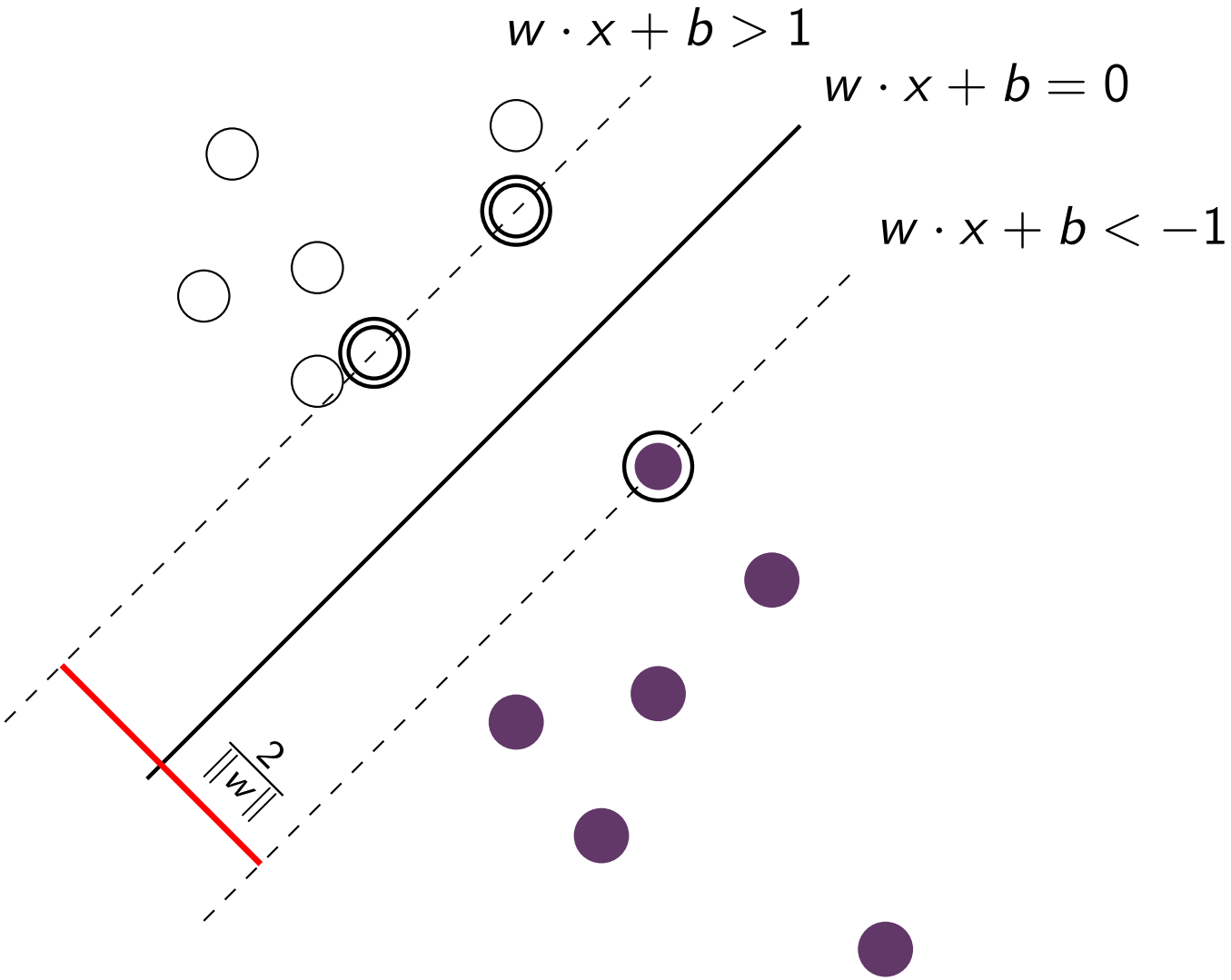
$$\text{maximize}_{w,b} \quad \frac{2}{\|w\|} \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\, i.$$

Equivalent formulation:

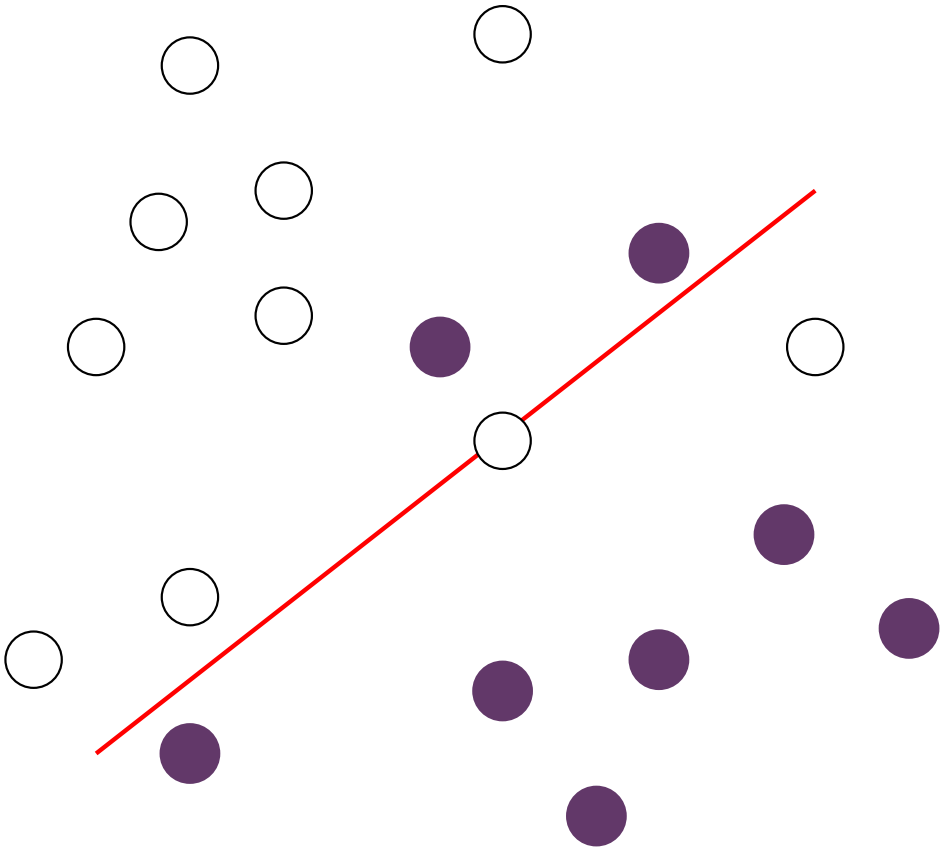
$$\text{minimize}_{w,b} \quad \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(x_i^\top w + b) \geq 1, \quad \forall\, i.$$

- We prefer to use $\|w\|^2$ instead of $\|w\|$ because $w \mapsto \|w\|$ is not differentiable at 0.
- Nonlinear (quadratic objective), constrained, continuous.

Support Vector Machines: Illustration



Support Vector Machines: Misclassification



- Question: What to do if the training set can not be perfectly separated by a hyperplane?

Handling Misclassification

Strategy and the Full SVM Problem:

$\rightsquigarrow$ Try to minimize the total or misclassification error:

$$\sum_{i=1}^{m} \max\{0, 1 - y_i \ell(x_i)\} \quad \text{(Hinge-Loss Function)}.$$

- SVM combines large margin and small misclassification cond.:

$$\min_{w,b} \quad \frac{\lambda}{2}\|w\|^2 + \sum_{i=1}^{m} \max\{0, 1 - y_i(x_i^\top w + b)\}, \quad \lambda > 0.$$

- λ is chosen to balance margin and misclassification. (Typically: $\lambda = \frac{1}{m}$ $\rightsquigarrow$ fine-tuning ...).
- This is an unconstrained, nonlinear, continuous problem.
- We now show how to equivalently rewrite it as a linear optimization problem in the case $\lambda = 0$.

A Linear Optimization Formulation for SVMs - I

Define $t_i = \max\{0, 1 - y_i(x_i^\top w + b)\} =: (1 - y_i(x_i^\top w + b))^+$.

We can first rewrite the SVM problem as follows:

$$\begin{aligned} \text{minimize}_{w,b,t} \quad & \sum_i t_i \\ \text{subject to} \quad & t_i = (1 - y_i(x_i^\top w + b))^+, \quad \forall\, i. \end{aligned}$$

We claim that we can relax "=" to "$\geq$" (why?):

$$\begin{aligned} \text{minimize}_{w,b,t} \quad & \sum_i t_i \\ \text{subject to} \quad & t_i \geq (1 - y_i(x_i^\top w + b))^+, \quad \forall\, i. \end{aligned}$$

A Linear Optimization Formulation for SVMs - II

Furthermore, $t_i \geq (1 - y_i(x_i^\top w + b))^+$ is equivalent to:

$$t_i \geq 1 - y_i(x_i^\top w + b), \qquad t_i \geq 0.$$

Therefore, the optimization problem can be reformulated as:

$$\begin{array}{lll} \text{minimize}_{w,b,t} & \sum_i t_i & \\ \text{subject to} & y_i(x_i^\top w + b) + t_i \geq 1, & \forall\ i \\ & t_i \geq 0 & \forall\ i. \end{array}$$

- This is a linear optimization problem with decision variables $w \in \mathbb{R}^n$, $b \in \mathbb{R}$, and $t \in \mathbb{R}^n$.

Questions?